Citizen Science in Galleries, Libraries, Archives, and Museums (GLAMs): Examining Inclusion in Digital Heritage Projects

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Galleries, libraries, archives and museums (GLAMs) preserve and share cultural heritage, providing access to the public and researchers (Bastian 2017). GLAMs have a shared aim to collect and make accessible knowledge in the public interest (Hemsworth / Castle 2022). GLAMs help us to understand and relate to culture, connecting the past with the future (Davis / Howard 2013). This may shape how visitors connect with their own personal identity (Packer 2008). They play a role as places of learning, imagination, personal development, and social interaction (Packer 2008).

Despite the role they play in our cultural heritage and society, GLAMs traditionally lack diverse audiences. In the UK, demographics of people who are most likely to engage with GLAMs digitally and/or physically are female, white, university-educated, from higher occupational classes, and without a disability (Mihelj et al. 2019). Similar trends are found in the USA, with frequent museum goers more likely to be white, over 60, and college-educated (American Alliance of Museums 2023).

Excluded groups miss the benefits of GLAM engagement, which fosters learning and deeper connections with culture and heritage (Seaward et al. 2022). GLAMs also lose out by not engaging diverse voices. Many collections contain objects taken during colonialism, and there are efforts to decolonise these collections through repatriation or other means of better representing the historical, political, and cultural significance of the collections (Hemsworth / Castle 2022). Engaging diverse audiences is crucial for this process. As GLAMs work in the public interest then they should be reflecting the range of identities and diverse communities that they represent (Hemsworth / Castle 2022).

GLAMs employ various strategies to overcome physical, cultural, and social barriers. Participatory approaches that bring the public into discussions and the work of GLAMs in various ways are popular methods to make GLAMs more inclusive and democratise heritage (Jansson 2023).

One example of public participation is Royal Botanic Gardens Kew's online digitization project, where members of the public are invited to transcribe handwritten specimen labels into a digital format. This type of project falls under Citizen Science, which RBG Kew defines as: "involvement of the public in scientific research to generate new knowledge or understanding" (Royal Botanic Gardens, Kew no date). Citizen science is not only a method to support digitisation of archive material, but also offers a way for audiences to deepen engagement with science and cultural heritage (Ridge 2013; Constantinidis 2016; Seaward et al. 2022). Natural history museums have an increasing role in supporting public education in biodiversity challenges, including reaching 'non-traditional' audiences, and citizen science has become a method for this (Sforzi et al. 2018). Through projects such as transcribing archives, people have the opportunity to engage with collections and their contents in different ways (Owens 2013).

However, research on citizen science participants has shown that these tend to be white, highly educated, have higher income, and employed or retired (Pateman et al. 2021). Given the overlap with demographics of those already engaging with GLAMs, it's unlikely that these interventions will organically reach a broader audience than those who already engage with these institutions (Bonacchi et al. 2019). Moreover, many institutions do citizen science online, which brings into question if inequalities in participation are further compounded by the 'digital divide': where those with higher socioeconomic status are further advantaged by digital technologies and those with lower socioeconomic status are further disadvantaged (Rogers 2001).

Many GLAMs lack data on digital audiences, making it hard to assess if they reach beyond traditional demographics. Research into digital strategies at four Oxford-based GLAMs found their digital audience data limited or non-existent (McCarthy 2019). Even with efforts to democratise heritage, participation may not translate into impact if volunteers are not representative of the broader public (Jansson 2023).

Given GLAMs' societal role and challenges in attracting diverse audiences, they must assess how effectively their engagement methods reach representative groups. This research, in partnership with RBG Kew, explores the role of citizen science in diversifying engagement through archive digi-

tisation at Kew and other institutions. The PhD research will examine current digital citizen science practices and recommend ways to design inclusive projects. The research questions are:

1. Why is citizen science used in GLAMs digitisation projects?
2. Who takes part in the projects and why?
3. What are the barriers and opportunities to design the projects more equitably?

This presentation will focus on research question one which draws on existing literature and interviews with project leaders to understand why citizen science is chosen for their projects, and specifically if the diversity of participation was a consideration.

Interviews were conducted with project leaders from 10 different archive digitisation projects across a range of locations, platforms and topics. The criteria for these was: they must take place online; be transcription-type projects (the main task for volunteers is transcribing text such as specimens or archives); be voluntary; be in English; and be currently active. Only voluntary projects were chosen as paid activities change the nature of engagement.

Interviews were chosen to give participants the space to give open-ended responses about their experiences of running citizen science projects and give richer data from the answers to analyse what themes come out of their responses (Braun / Clarke 2013). Each interviewee was asked the same set of standard questions from an interview guide, with the ability to ask follow up questions to probe further on subjects of interest. Combining the use of an interview guide with standardised questions allowed for flexibility to explore certain topics in greater depth while keeping consistency across interviews (Patton 2015).

The presentation will summarise the thematic analysis of the interviews, covering why citizen science was used, what the aims of engagement were, and whether or not inclusion is a driving force behind the use of the approach. This analysis will give an insight into why GLAMs choose citizen science to support their archive digitisation and form the basis of further research into how to make these projects inclusive.

Bibliography

American Alliance of Museums (2023): “Museum Visitation—Frequency vs. Incidence Gaps: A 2023 Annual Survey of Museum-Goers Data Story”, *American Alliance of Museums*. <https://www.aam-us.org/2023/11/17/museum-visitation-frequency-vs-incidence-gaps-a-2023-annual-survey-of-museum-goers-data-story/> [31.05.2024].

Bastian, Jeannette (2017): “GLAMs, LAMs, and Archival Perspectives”, in: *Currents of Archival Thinking*, Libraries Unlimited. 327–350.

Bonacchi, Chiara / Bevan, Andrew / Keinan-Schoonbaert, Adi / Pett, Daniel / Wexler, Jennifer (2019): “Participation in heritage crowdsourcing”, in: *Museum Management and Curatorship* 34, 2: 166–182. DOI: 10.1080/09647775.2018.1559080.

Braun, Virginia / Clarke, Victoria (2006): “Using thematic analysis in psychology”, in: *Qualitative Research in Psychology* 3, 2: 77–101. DOI: 10.1191/1478088706qp063oa.

Braun, Virginia / Clarke, Victoria (2013): *Successful Qualitative Research*. London: Sage Publications.

Constantinidis, Dora (2016): “Crowdsourcing Culture: Challenges to Change”, in: Borowiecki, Karol Jan / Forbes, Neil / Fresa, Antonella (eds): *Cultural Heritage in a Changing World*. Cham: Springer International Publishing 215–234. DOI: 10.1007/978-3-319-29544-2_13.

Davis, Wendy / Howard, Katherine (2013): “Cultural policy and Australia’s national cultural heritage: issues and challenges in the GLAM landscape”, in: *The Australian Library Journal* 62, 1: 15–26. DOI: 10.1080/00049670.2013.774684.

Sforzi, Andrea / Tweddle, John / Vogel, Johannes / Lois, Grégoire / Wägele, Wolfgang / Lakeman-Fraser, Poppy / Makuch, Zen / Vohland, Katrin (2018): “Citizen science and the role of natural history museums”, in: Hecker, Susanne / Haklay, Muki / Bowser, Anne / Makuch, Zen / Vogel, Johannes / Bonn, Aletta (eds.): *Citizen Science Innovation in Open Science*. London: UCL Press. 429-444 <https://doi.org/10.14324 /111.9781787352339> [27.08.2024].

Hemsworth, Katie / Castle, Ysabel (2022): “Historical Research and Storytelling Through “GLAMs”, in: *Spatial Humanities and Digital Storytelling: Critical Historical Approaches.* <https://ecampusontario.pressbooks.pub/spatialhumanities/chapter/historical-research-and-storytelling-through-glams-2/> [12.11.2024].

Jansson, Ina-Maria (2023): “Challenging the problem of un-democratic participation: from destruction to re-construction of heritage”, in: *Journal of Documentation* 79, 2: 509–526. DOI: 10.1108/JD-01-2022-0025.

McCarthy, Elizabeth (2019): “Developing digital audience frameworks for Oxford’s GLAM”, in: *Performance Measurement and Metrics*, 20, 3: 179–185.

Mihelj, Sabina / Leguina, Adrian / Downey, John (2019) “Culture is digital: Cultural participation, diversity and the digital divide”, in: *New Media & Society* 21, 7: 1465–1485. DOI: 10.1177/1461444818822816.

Owens, Trevor (2013): “Digital Cultural Heritage and the Crowd”, in: *Curator: The Museum Journal* 56, 1: 121–130. <https://doi.org/10.1111/cura.12012> [27.08.2024].

Packer, Jan (2008): “Beyond Learning: Exploring Visitors’ Perceptions of the Value and Benefits of Museum Experiences”, in: *Curator: The Museum Journal* 51, 1: 33–54. <https://doi.org/10.1111/j.2151-6952.2008.tb00293.x.> [13.11.2024].

Pateman, Rachel / Dyke, Alison / West, Sarah (2021): “The Diversity of Participants in Environmental Citizen Science”, in: *Citizen Science: Theory and Practice* 6, 1: 9.

Patton, Michael Quinn (2015): *Qualitative research & evaluation methods: integrating theory and practice*. SAGE PUBLICATIONS.

Ridge, Mia (2013): “From Tagging to Theorizing: Deepening Engagement with Cultural Heritage through Crowdsourcing”, in: *Curator: The Museum Journal* 56, 4: 435–450. <https://doi.org/10.1111/cura.12046.> [25.04.2024].

Rogers, Everett M. (2001): “The Digital Divide”, in: *Convergence* 7, 4: 96–111. <https://doi.org/10.1177/135485650100700406.> [28.01.2024]

Royal Botanic Gardens, Kew (2022): “Outreach Strategy 2022–2027”. *Royal Botanic Gardens, Kew*. < https://www.kew.org/sites/default/files/2022-11/Kew-Outreach-Strategy-2022-2027.pdf> [26.04.2024]

Seaward, Louise / Willcox, Pip / Blickhan, Samantha / Bligh, Stuart / Butler, Will / Fulton, Liz / Haston, Elspeth / Hawkins, Ashleigh / Hutcheon, Rebecca / King, Sally / Kocsis, Andrea / Lintott, Chris / Miller, Grant / Nash, Trevor / O'Donnell, Jim / Ogden, Bernard / Salmon, Martin / Smith, Thomasina (2022): “Engaging Crowds: citizen research and cultural heritage data at scale”, *Zenodo*. <https://doi.org/10.5281/zenodo.7152031.> [01.11.2024].